

analytical processing) analysis. It has good alignment with the rest of the UIDAI stack.

Apache Tomcat is used as a Web container wherever Http access of dynamic data is needed.

MongoDB: The distributed document oriented database provides schema flexibility, low latency data reads, and auto-sharding of content.

Apache Solr is used for fast search index of full text data.

MySQL is the RDBMS store for all relational data that requires ACID (atomicity, consistency, isolation and durability) properties, which is heavily sharded for scaling.

Apart from this, the project makes use of the Spring framework, an application container and custom-built server-like runtime for configuration, administration, call interception, monitoring and metrics collection. "We also use many of its sub-projects like Spring security, Spring batch, Spring MVC (model-view-controller), etc.," reveals Balasubramanian.

The biometric devices and scanners used for enrolment are procured from the market by enrolment agencies. Initially, most vendors supported device drivers only for Windows. Now, drivers are available for other operating systems like Linux. The enrolment client software works flawlessly with these drivers on non-Windows operating systems as well, Balasubramanian affirms.

Since the data gathered for the UIDAI project is sensitive and is an access point for many different benefits, it also requires backups in case of crashes or physical corruption of data. So how does the UIDAI team ensure data security and disaster management? Balasubramanian divulges more details: "The UIDAI has a comprehensive security policy to ensure the safety and integrity of its data. Data in transit is encrypted to protect against tampering. The Aadhaar datastores are secured both physically and electronically, and access is given only to a few select individuals. Data is secured with the best encryption and in a highly secure data vault. All access details are logged. There are strict data retention guidelines that are adhered to, including rules related to redundant copies and multi-data centre deployment. We use FOSS distributed file systems (with data replication) for realising high throughput data read-writes within the data centre."

A few 'open' practices

Balasubramanian has been with the Aadhaar project from the time the work to build the core application was awarded to MindTree. He has since led the architecture, design and development efforts on the project. His team has built the enrolment client, enrolment and authentication servers, the portals and the business intelligence platform for the project. He shares a few best practices that have helped him and his team in their journey with the project so far: "Some of the best practices that we follow have been driven in part by the chief architect of the project, Dr Pramod Varma. For example, we follow agile development so that feature releases can happen more often and feedback can be incorporated. There is a strong design focus where no part of the system is stereotyped to a certain product or approach. Fresh thinking is encouraged, and we try to keep our

Aadhaar at Open Source India 2011



Regunath Balasubramanian,
principal architect of the UIDAI
project
session and many others will shortly be uploaded on
osidays.com and [Linuxforu.com](http://linuxforu.com).

Regunath Balasubramanian, principal architect, the UIDAI project, an eminent speaker and guest of honour at the recently concluded Open Source India 2011 convention, shared with the audiences the complexities of a project as huge as Aadhaar and explained in detail the advantages that the OSS stack has rendered to the project. Glimpses from this project. Glimpses from this session and many others will shortly be uploaded on osidays.com and Linuxforu.com.

design principles simple and follow patterns whenever prudent."

An 'open' platform for future e-governance projects

The Aadhaar project is consuming a lot of FOSS. The general interest in the project and in nation building has invited contributions and support from volunteers, user communities and technology teams, in general. "This collective wisdom has helped in feature design, technology selection and making course corrections, especially when solutions don't work as required," reveals Balasubramanian.

But is the project team also contributing back to the community? Balasubramanian affirms: "As far as giving back to the community goes, the UIDAI team interacts with user and developer communities of certain solutions that are being used in the project, the more significant being RabbitMQ, Hadoop and MongoDB. The developer community of RabbitMQ has been particularly helpful in our early days of adoption and, more recently, it has been the MongoDB community."

Apart from this, the e-governance platform that the team has built is intended for use by other e-governance initiatives in India, informs Balasubramanian. "I would recommend the use of FOSS and, in fact, the reuse of the e-governance platform that we have built for the UIDAI project. We have written a few technology components as part of the platform that can integrate with existing frameworks. These may be open sourced when the time is appropriate and sustenance is taken care of," he adds.

Looking at this success story and considering how well FOSS has worked for the Aadhaar project so far, we are sure many other projects and initiatives will be inspired to leverage FOSS and enjoy many of its benefits. **END** 

By: Vandana Sharma

The author is executive editor of LFY. Apart from writing on subjects like entrepreneurship and IT for businesses, she loves to read on a range of topics, which include religion and spirituality. She enjoys 'meaningful' fiction, too.